

Day 03 – Enterprise DNS & DHCP Integration Lab

Author: Theo Mata

Track: Networking / Cybersecurity / Linux Administration

Platform: Proxmox VE

Lab Type: Enterprise Infrastructure Services

Focus Areas: DNS, DHCP, OPNsense, BIND9, Troubleshooting, Network Services

Overview

The goal of Day 03 was to build an enterprise-style DNS infrastructure and integrate it with centralized DHCP services.

Unlike home networks where DNS and DHCP are often provided by a single router, enterprise environments typically separate services into dedicated servers.

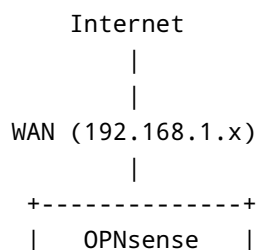
In this lab:

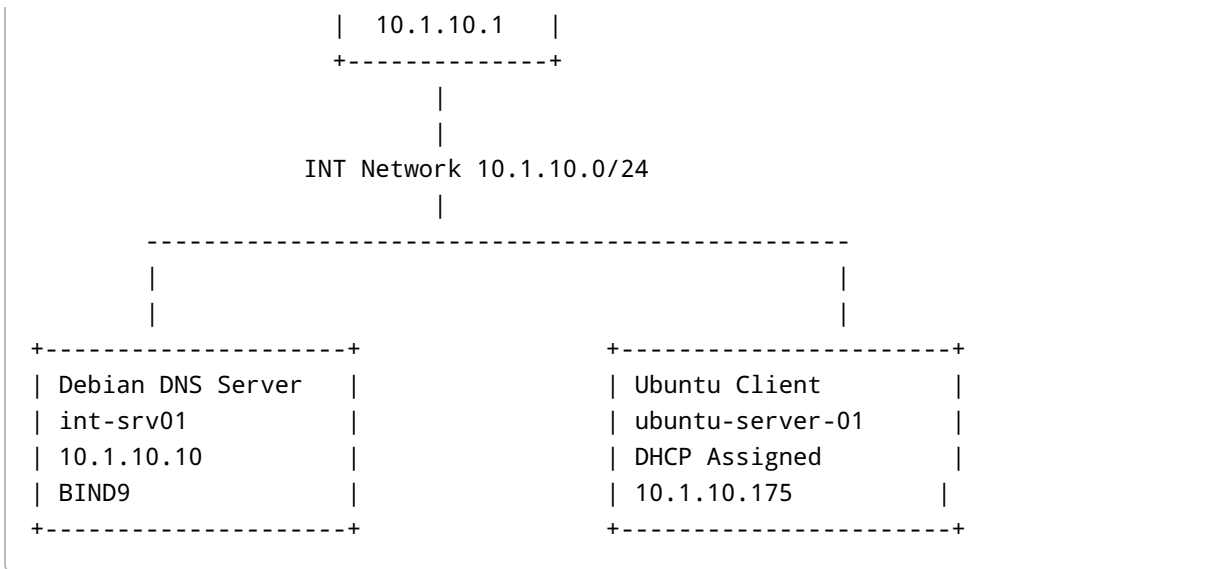
- OPNsense acts as the router and DHCP server
- Debian acts as the dedicated DNS server
- Ubuntu acts as a DHCP client
- Proxmox acts as the virtualization platform

The final objective was to allow DHCP clients to:

- Automatically receive an IP address
 - Automatically receive a default gateway
 - Automatically receive a DNS server
 - Automatically receive a domain search suffix
 - Resolve internal hostnames
 - Resolve external internet domains
-

Final Network Topology





Infrastructure Components

Proxmox VE

Purpose:

- Hypervisor
- VM management
- Virtual networking

OPNsense

Purpose:

- Router
- Firewall
- Gateway
- DHCP Server

IP Address:

10.1.10.1

Debian 12

Hostname:

int-srv01

IP Address:

10.1.10.10

Purpose:

Authoritative DNS Server

Software:

BIND9

Ubuntu Server

Purpose:

DHCP Client Testing
DNS Client Testing

IP Address:

10.1.10.175

Assigned dynamically by DHCP.

OPNsense Installation

A dedicated OPNsense virtual machine was created in Proxmox.

Virtual interfaces were attached:

```
WAN
LAN
OPT1
OPT2
```

Detected inside OPNsense as:

```
vtnet0
vtnet1
vtnet2
vtnet3
```

Interface Assignment

Configured:

```
WAN  -> vtnet0
LAN  -> vtnet1
OPT1 -> vtnet2
OPT2 -> vtnet3
```

Result:

```
WAN  -> 192.168.1.x
LAN  -> 10.1.10.1/24
OPT1 -> 10.1.20.1/24
OPT2 -> 10.1.40.1/24
```

Connectivity Validation

Verified internet connectivity from OPNsense.

Test:

```
ping 8.8.8.8
```

Result:

```
0% packet loss
```

DNS validation:

```
ping google.com
```

Result:

```
Successful
```

This confirmed:

- Routing works
 - Gateway works
 - WAN works
 - Internet access works
-

DNS Server Deployment

A Debian 12 server was provisioned.

Verification:

```
hostnamectl
```

Output:

```
Static hostname: int-srv01

Operating System:
Debian GNU/Linux 12 (bookworm)

Kernel:
Linux 6.1
```

Configure Static IP Address

DNS servers should always have static IP addresses.

Configured:

```
10.1.10.10/24
```

Gateway:

```
10.1.10.1
```

DNS:

```
10.1.10.1
```

BIND9 Installation

Installed required packages:

```
apt update  
  
apt install bind9 bind9-utils dnsutils -y
```

Packages:

```
bind9  
bind9-utils  
dnsutils
```

Purpose:

bind9

```
DNS Server
```

dnsutils

```
dig
nslookup
host
```

bind9-utils

```
named-checkconf
named-checkzone
```

Verify Service Status

Command:

```
systemctl status named
```

Expected:

```
active (running)
```

Observed:

```
active (running)
```

Service successfully started.

Verify Listening Ports

Command:

```
ss -tulnp | grep named
```

Observed:

TCP 53
UDP 53

Listening on:

10.1.10.10
127.0.0.1
IPv6 addresses

This confirmed BIND is accepting DNS queries.

DNS Configuration

File:

/etc/bind/named.conf.local

Configuration:

```
zone "worldskills.ph" {  
    type master;  
    file "/etc/bind/db.worldskills.ph";  
};  
  
zone "10.1.10.in-addr.arpa" {  
    type master;  
    file "/etc/bind/db.10.1.10";  
};
```

Understanding the Zones

Forward Zone:

worldskills.ph

Used to translate:

Hostname -> IP Address

Example:

int-srv01.worldskills.ph

↓

10.1.10.10

Reverse Zone:

10.1.10.in-addr.arpa

Used to translate:

IP Address -> Hostname

Example:

10.1.10.10

↓

int-srv01.worldskills.ph

Forward Zone File

Created:

/etc/bind/db.worldskills.ph

Contents:

```
$TTL 604800
```

```
@ IN SOA ns1.worldskills.ph. admin.worldskills.ph. (
    2026052901
    604800
    86400
    2419200
    604800 )
```

```
@ IN NS ns1.worldskills.ph.
```

```
ns1          IN A    10.1.10.10
```

```
int-srv01    IN A    10.1.10.10
```

```
@            IN A    10.1.10.10
```

Record Breakdown

SOA

Start of Authority

Identifies:

- Primary DNS Server
- Administrator
- Zone serial number

NS

Name Server Record

Defines:

ns1.worldskills.ph

as authoritative DNS server.

A Record

Maps:

Hostname -> IPv4 Address

Example:

int-srv01



10.1.10.10